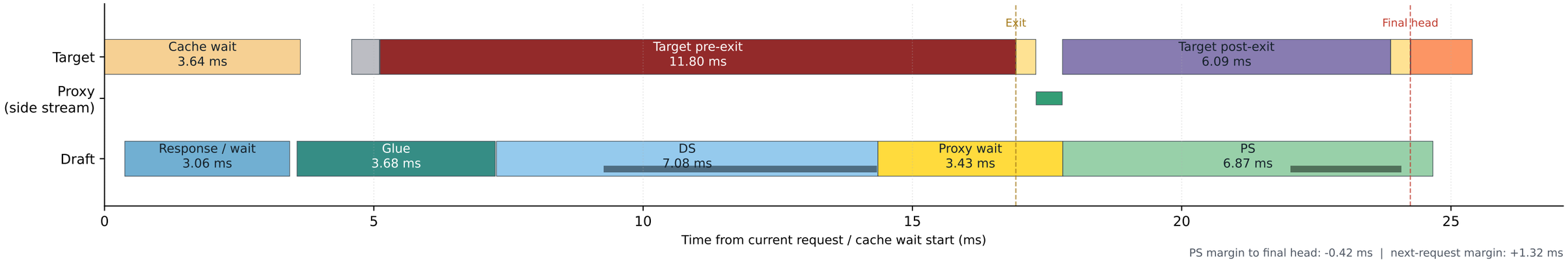
LayerSkip Llama2-7B + llama-68M
Target / draft aligned timeline

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 82d964e0 / eslab18 RTX3090 / B1, T0.7 / 4 profiled turns, output cap256 / 321 eligible steps

P1 hit | actual p50 step / n=115 | UID 244_t0, step 78 | AL=3

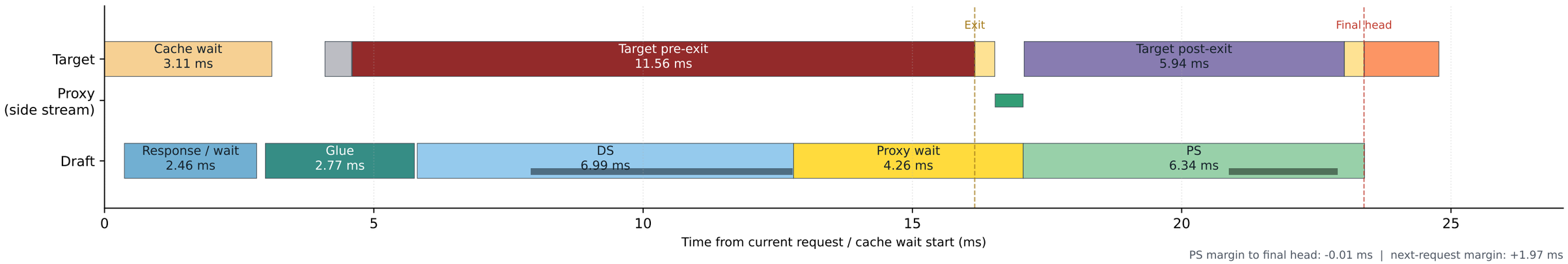
DS 7.08 ms / PS 6.87 ms



PS margin to final head: -0.42 ms | next-request margin: +1.32 ms

P2 hit | actual p50 step / n=164 | UID 84_t0, step 96 | AL=3

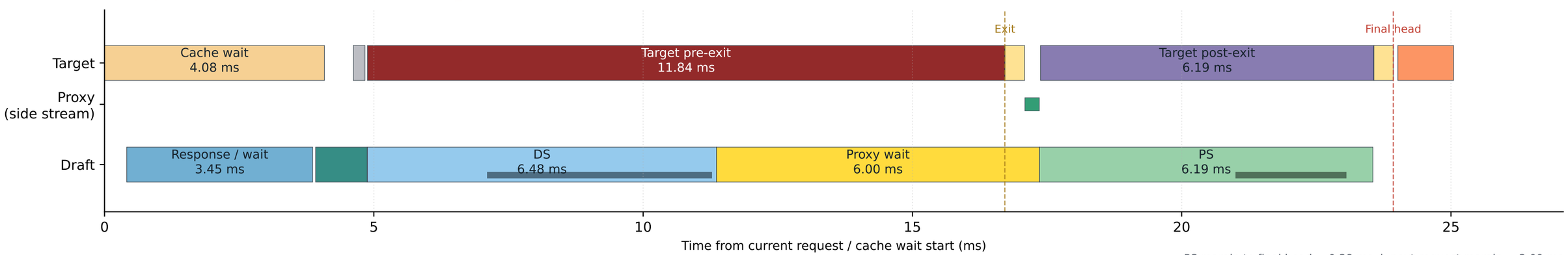
DS 6.99 ms / PS 6.34 ms



PS margin to final head: -0.01 ms | next-request margin: +1.97 ms

Miss | actual p50 step / n=42 | UID 288_t0, step 84 | AL=1

DS 6.48 ms / PS 6.19 ms



PS margin to final head: +0.38 ms | next-request margin: +2.09 ms



Dark stripe inside DS/PS = graph replay. Full bar includes preparation / export / merge. Real measured steps, no clock shift. Blank gaps are retained.

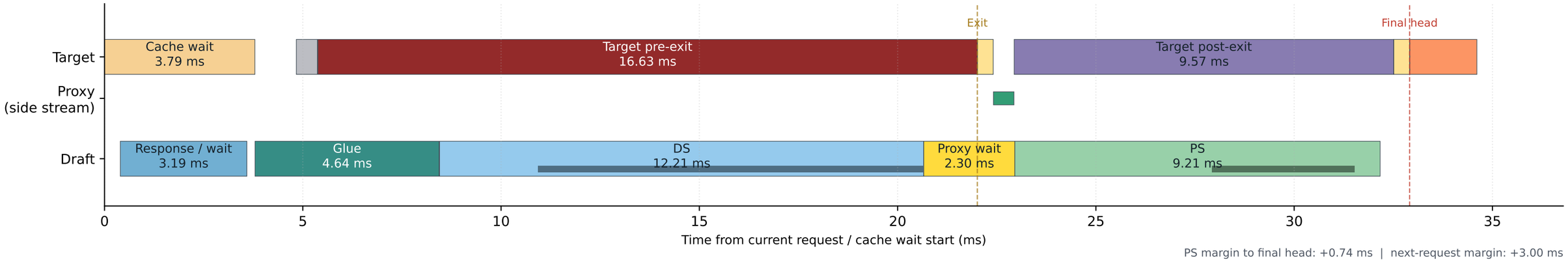
LayerSkip Llama2-13B + AMD-Llama-135M
Target / draft aligned timeline

exit 23 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | last | target GPUs=2 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 59f46511 / eslab18 RTX3090 / B1, T0.7 / 4 profiled turns, output cap256 / 444 eligible steps

P1 hit | actual p50 step / n=139 | UID 84_t0, step 68 | AL=5

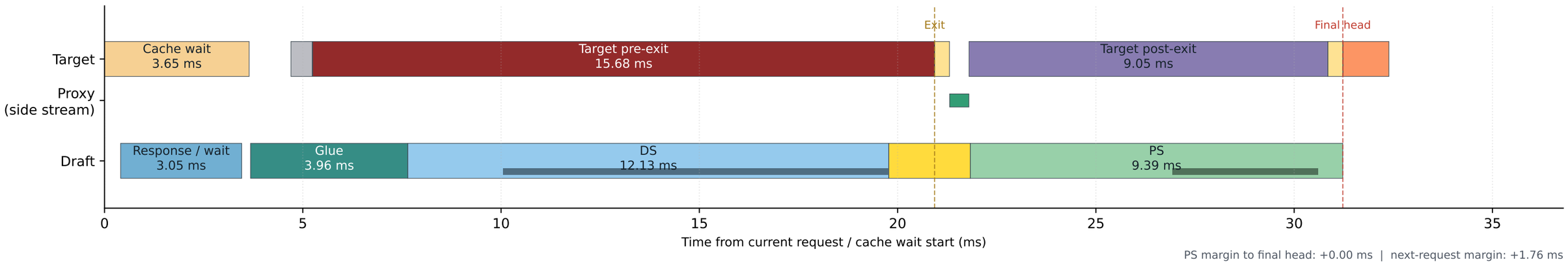
DS 12.21 ms / PS 9.21 ms



PS margin to final head: +0.74 ms | next-request margin: +3.00 ms

P2 hit | actual p50 step / n=225 | UID 288_t0, step 127 | AL=2

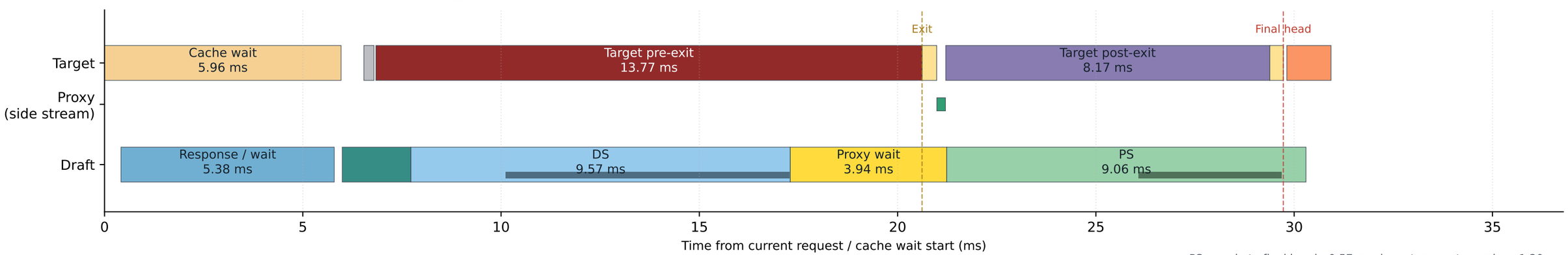
DS 12.13 ms / PS 9.39 ms



PS margin to final head: +0.00 ms | next-request margin: +1.76 ms

Miss | actual p50 step / n=80 | UID 128_t0, step 29 | AL=1

DS 9.57 ms / PS 9.06 ms



PS margin to final head: -0.57 ms | next-request margin: +1.20 ms

Cache wait | Prepare | Target pre-exit | Target post-exit | LM head | Acceptance | Proxy | Response / wait | Glue | DS | Proxy wait | PS | Previous PS tail

Dark stripe inside DS/PS = graph replay. Full bar includes preparation / export / merge. Real measured steps, no clock shift. Blank gaps are retained.

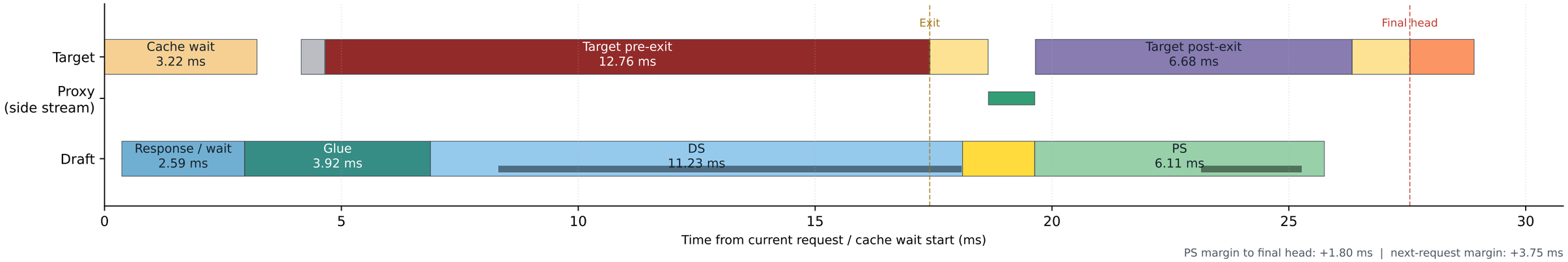
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Target / draft aligned timeline

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 16dd4198 / eslab18 RTX3090 / B1, T0.7 / 4 profiled turns, output cap256 / 387 eligible steps

P1 hit | actual p50 step / n=65 | UID 84_t0, step 170 | AL=4

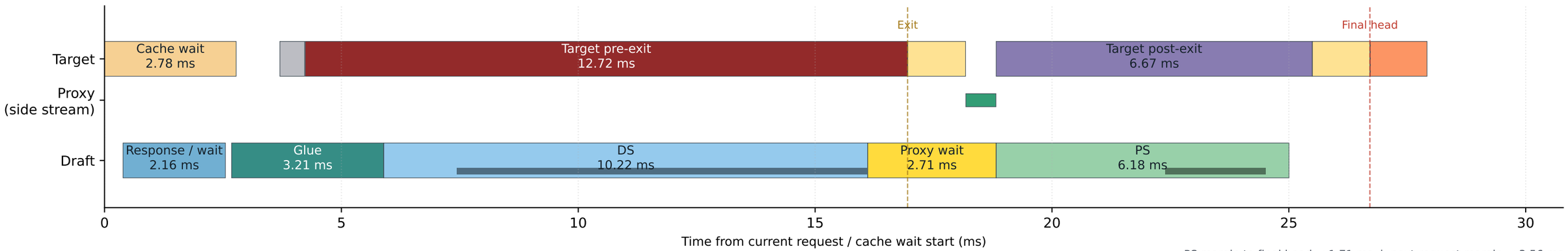
DS 11.23 ms / PS 6.11 ms



PS margin to final head: +1.80 ms | next-request margin: +3.75 ms

P2 hit | actual p50 step / n=295 | UID 244_t0, step 92 | AL=1

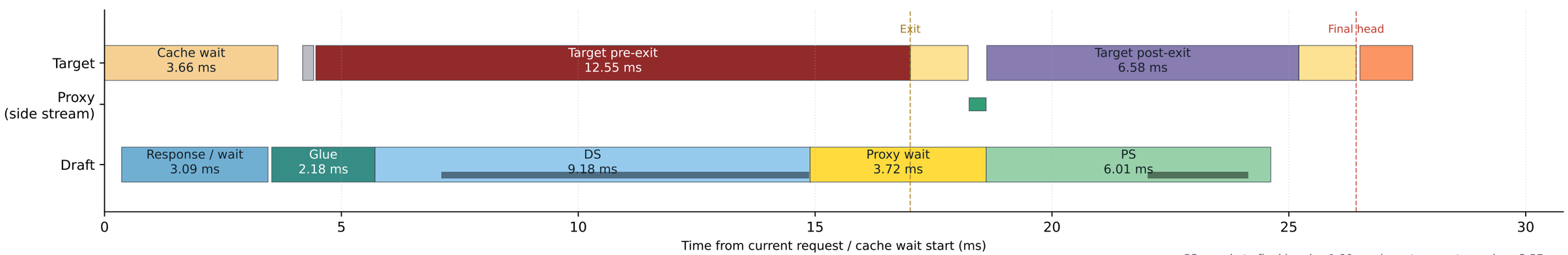
DS 10.22 ms / PS 6.18 ms



PS margin to final head: +1.71 ms | next-request margin: +3.56 ms

Miss | actual p50 step / n=27 | UID 244_t0, step 65 | AL=1

DS 9.18 ms / PS 6.01 ms



PS margin to final head: +1.80 ms | next-request margin: +3.57 ms

Cache wait | Prepare | Target pre-exit | Target post-exit | LM head | Acceptance | Proxy | Response / wait | Glue | DS | Proxy wait | PS | Previous PS tail

Dark stripe inside DS/PS = graph replay. Full bar includes preparation / export / merge. Real measured steps, no clock shift. Blank gaps are retained.

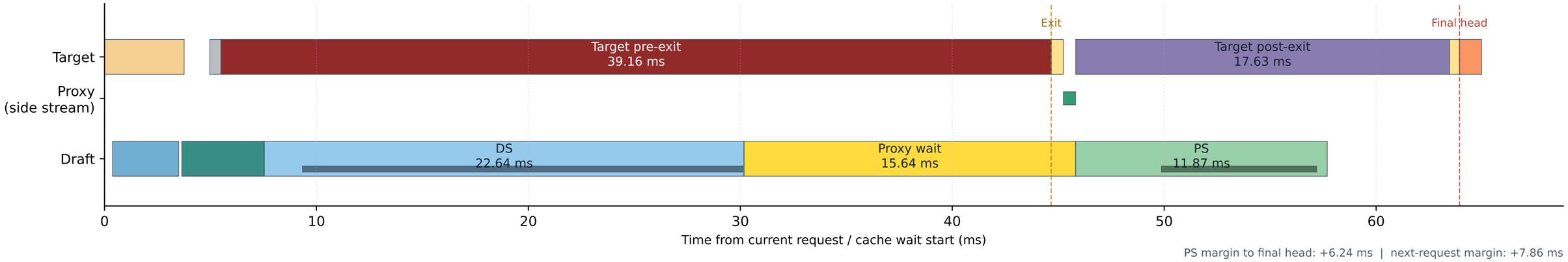
LayerSkip CodeLlama-34B + AMD-Llama-135M-code
Target / draft aligned timeline

exit 30 (0-based) | K1/K2 12/6 | G=M DS/PS 12/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 59f46511 / eslab18 RTX3090 / B1, T0.7 / 4 profiled turns, output cap256 / 73 eligible steps

P1 hit | actual p50 step / n=62 | UID humaneval_local_033_t0, step 7 | AL=4

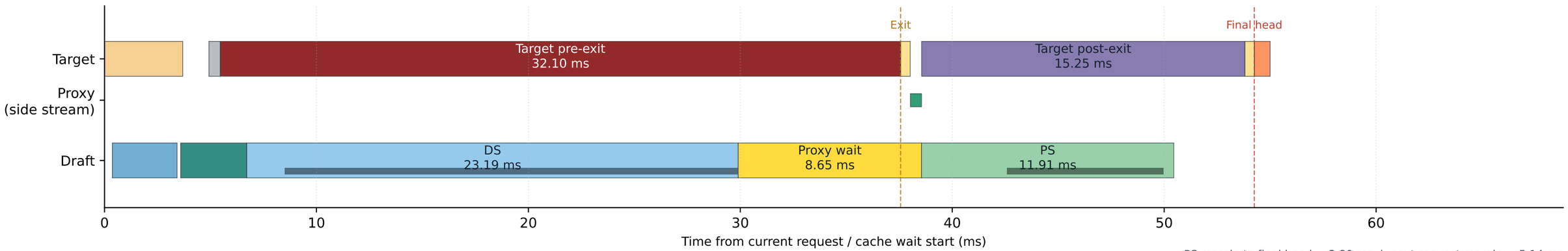
DS 22.64 ms / PS 11.87 ms



PS margin to final head: +6.24 ms | next-request margin: +7.86 ms

P2 hit | actual p50 step / n=11 | UID humaneval_local_023_t0, step 25 | AL=2

DS 23.19 ms / PS 11.91 ms



PS margin to final head: +3.80 ms | next-request margin: +5.14 ms

Miss — no eligible samples

Cache wait | Prepare | Target pre-exit | Target post-exit | LM head | Acceptance | Proxy | Response / wait | Glue | DS | Proxy wait | PS | Previous PS tail

Dark stripe inside DS/PS = graph replay. Full bar includes preparation / export / merge. Real measured steps, no clock shift. Blank gaps are retained.

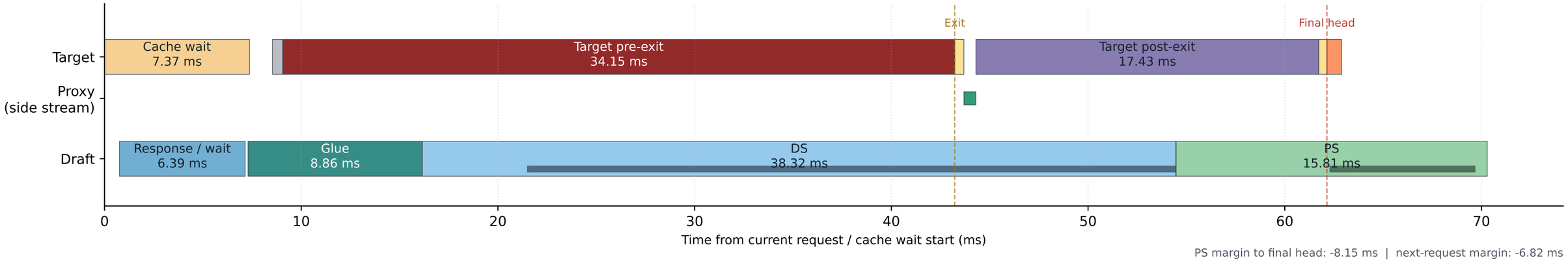
CodeLlama-34B + AMD-135M-code / throughput alternative
Target / draft aligned timeline

exit 30 (0-based) | K1/K2 16/6 | G=M DS/PS 16/8 | C=3 | last | target GPUs=4 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 59f46511 / eslab18 RTX3090 / B1, T0.7 / 4 profiled turns, output cap256 / 84 eligible steps

P1 hit | actual p50 step / n=63 | UID humaneval_local_013_t0, step 30 | AL=2

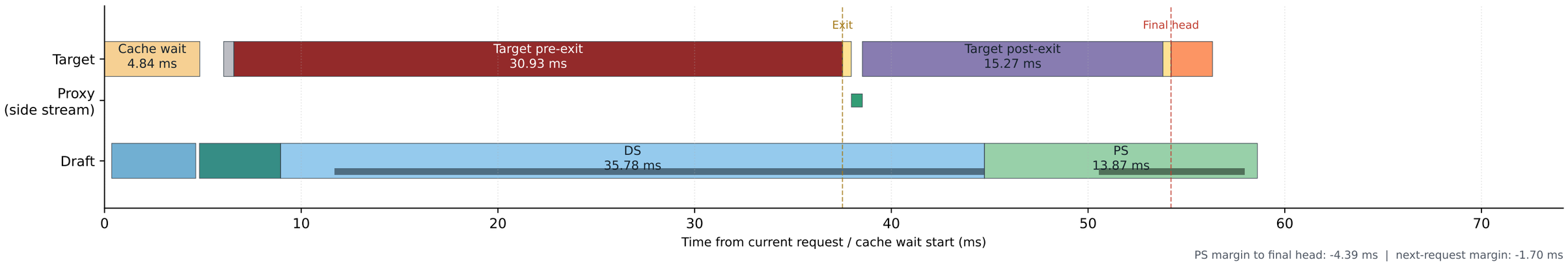
DS 38.32 ms / PS 15.81 ms



PS margin to final head: -8.15 ms | next-request margin: -6.82 ms

P2 hit | actual p50 step / n=19 | UID humaneval_local_033_t0, step 11 | AL=7

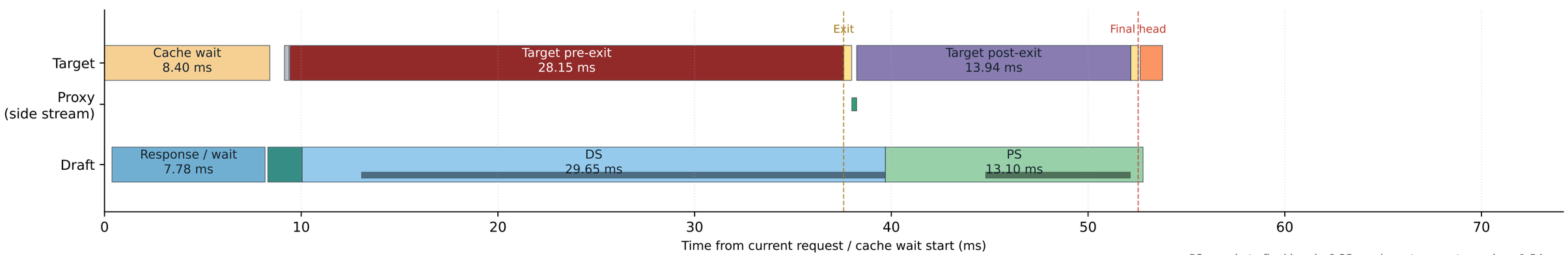
DS 35.78 ms / PS 13.87 ms



PS margin to final head: -4.39 ms | next-request margin: -1.70 ms

Miss | actual p50 step / n=2 | UID humaneval_local_023_t0, step 6 | AL=1

DS 29.65 ms / PS 13.10 ms



PS margin to final head: -0.25 ms | next-request margin: +1.54 ms

Cache wait | Prepare | Target pre-exit | Target post-exit | LM head | Acceptance | Proxy | Response / wait | Glue | DS | Proxy wait | PS | Previous PS tail

Dark stripe inside DS/PS = graph replay. Full bar includes preparation / export / merge. Real measured steps, no clock shift. Blank gaps are retained.